COURSE 4 · CHEMISTRY

Chemistry

Theory reader, audio edition

Atoms, the periodic table, bonding, the mole, gases, reactions, equilibrium, acids and redox, and the organic chemistry of life.

8 modules · equations spoken in words; numbered equations appear in the Visual Companion.

Before you start

How to use this reader

This is the audio edition. It is written to be listened to in a text-to-speech app as comfortably as read on the page. Equations are spoken in words, and each important one carries a number like this: [Equation 4.1-1]. The same number appears in the Visual Companion, where the equation is printed in ordinary notation.

Each module takes about twenty to thirty minutes to listen to. A good weekly rhythm at four hours: listen on the first day, look over the Visual Companion with a pencil, try the explorer, then spend the remaining time in the workbook.

The level sits between final-year high school and first-year university. Nothing is assumed beyond what you met at school, and it is fine if that was a long time ago. Where a first-year idea appears, it is introduced from the ground up.

In this course

4.1 Matter and atomic structure
4.2 The periodic table
4.3 Bonding and molecular shape
4.4 The mole and stoichiometry
4.5 States of matter and gas laws
4.6 Energy, rates and equilibrium
4.7 Acids, bases and redox
4.8 Organic chemistry essentials

Chemistry explains how atoms combine and rearrange to make everything around us. This course starts inside the atom, builds up to bonds and molecules, learns to count atoms by weighing them, and finishes with the reactions that power batteries, digest food and build living things. It uses the physics of energy from Course 3 and the logarithms of Course 2.

Module 4.1

Matter and atomic structure

Everything is made of atoms, and every atom is made of the same three particles. How many of each it has decides what element it is, how heavy it is, and how it behaves.

By the end of this module you can

- Describe protons, neutrons and electrons and their arrangement
- Use atomic number, mass number and isotopes
- Write electron configurations for the first twenty elements

Inside the atom

An atom has a tiny, dense nucleus of protons, which are positive, and neutrons, which are neutral. Around it is a cloud of electrons, which are negative and about two thousand times lighter. If the nucleus were a marble at the centre of a football stadium, the electrons would be specks in the outer stands. Almost all of an atom is empty space.

The atomic number Z is the number of protons, and it defines the element: every carbon atom has six. The mass number A is protons plus neutrons [Equation 4.1-1]. A neutral atom has as many electrons as protons. Ions have gained or lost electrons and so carry a charge.

Isotopes and relative atomic mass

Atoms of the same element can have different numbers of neutrons. These isotopes behave almost identically in chemistry but differ in mass and sometimes in stability. Carbon-twelve has six neutrons; carbon-fourteen has eight and is radioactive, as you met in Module 3.9.

The relative atomic mass on the periodic table is the weighted average over naturally occurring isotopes [Equation 4.1-2]. Chlorine is about seventy-five per cent chlorine-thirty-five and twenty-five per cent chlorine-thirty-seven, giving thirty-five point five.

Electron shells

Electrons occupy shells at increasing energies. A simple model, good up to calcium, puts at most two electrons in the first shell and eight in each of the next two. Sodium, with eleven electrons, has the arrangement two, eight, one. The electrons in the outermost shell, the valence electrons, decide how an element reacts.

First-year chemistry refines this into subshells labelled s, p, d and f, written for sodium as one s two, two s two, two p six, three s one [Equation 4.1-3]. The underlying reason is quantum mechanics: each subshell is a different pattern of electron wave, with room for a fixed number of electrons.

Key ideas to carry forward

- Z = protons (defines the element); A = protons + neutrons
- Isotopes differ in neutron number only
- Valence electrons decide chemistry
- Shells fill 2, 8, 8 for the first 20 elements

Module 4.2

The periodic table

The periodic table is chemistry's map. Arranged by atomic number, elements fall into columns with similar behaviour, and the trends across and down it let you predict properties of elements you have never seen.

By the end of this module you can

- Relate groups and periods to electron arrangement
- Describe trends in atomic radius, ionisation energy and electronegativity
- Describe the main families: alkali metals, halogens, noble gases and transition metals

Groups and periods

Dmitri Mendeleev published his table in 1869, leaving gaps for elements not yet discovered and predicting their properties. When gallium and germanium turned up matching his predictions, the table was vindicated.

Rows are periods: each new row starts a new electron shell. Columns are groups: elements in the same group have the same number of valence electrons, and so react similarly. Group one, the alkali metals, all have one outer electron and react vigorously with water. Group seventeen, the halogens, are one electron short of a full shell and grab electrons eagerly. Group eighteen, the noble gases, have full shells and barely react at all.

Trends

Across a period, protons are added to the nucleus while electrons go into the same shell. The stronger pull shrinks the atom, raises the energy needed to remove an electron, called ionisation energy, and increases electronegativity, the pull an atom exerts on shared electrons. Down a group, each new shell puts the outer electrons further from the nucleus and shields them, so atoms get bigger and their outer electrons easier to remove.

These trends explain reactivity. Alkali metals get more reactive down the group, because the outer electron is lost more easily: lithium fizzes in water, caesium explodes. Halogens get less reactive down the group, because gaining an electron becomes harder: fluorine is the most reactive element of all.

Metals and non-metals

Metals sit to the left and centre; they lose electrons, conduct electricity and form positive ions. Non-metals sit to the upper right; they gain or share electrons. The transition metals in the middle block form coloured compounds and make good catalysts, which matters for industry and for life: iron carries oxygen in your blood.

Key ideas to carry forward

- Group = valence electrons = similar chemistry
- Period = number of shells
- Across a period: smaller atoms, higher ionisation energy and electronegativity
- Down a group: larger atoms, outer electrons easier to remove

Module 4.3

Bonding and molecular shape

Atoms bond to reach more stable electron arrangements. The kind of bond, ionic, covalent or metallic, and the shape of the resulting molecule explain why salt dissolves, why water is liquid, and why diamond is hard.

By the end of this module you can

- Describe ionic, covalent and metallic bonding
- Predict molecular shapes with VSEPR and identify polar molecules
- Explain intermolecular forces, including hydrogen bonding

Three kinds of bond

In ionic bonding, a metal gives electrons to a non-metal, forming oppositely charged ions held in a lattice. Table salt is the classic case: hard, high melting, conducting only when molten or dissolved. In covalent bonding, non-metals share pairs of electrons; each shared pair is a bond, and atoms often share enough to fill their outer shells. In metallic bonding, metal atoms release outer electrons into a shared sea, which is why metals conduct and can be bent without shattering.

Bonds sit on a spectrum. When two atoms with different electronegativities share electrons, the sharing is unequal and the bond is polar, with a small negative charge on the more electronegative atom. Water's oxygen pulls electrons away from its hydrogens.

Molecular shape

Electron pairs around a central atom repel each other and spread out as far as possible. This is valence shell electron pair repulsion, VSEPR. Two pairs give a straight line, three a flat triangle, four a tetrahedron with angles of about one hundred and nine degrees. Lone pairs, which are not shared, repel more strongly and squash the bond angles. Water has two bonds and two lone pairs, so it is bent at about one hundred and four point five degrees.

Shape decides polarity. Carbon dioxide has polar bonds, but it is straight, so they cancel and the molecule is non-polar. Water's bent shape means its polar bonds do not cancel, so water is polar, with a positive end and a negative end.

Forces between molecules

Intermolecular forces hold molecules to one another. They are much weaker than bonds, but they decide boiling points and solubility. All molecules attract weakly through temporary electron fluctuations, called dispersion forces. Polar molecules also attract through their permanent charges. The strongest is the hydrogen bond, when hydrogen attached to nitrogen, oxygen or fluorine is attracted to a lone pair on another such atom.

Hydrogen bonding is why water boils at one hundred degrees when similar-sized molecules are gases, why ice floats, and why DNA's two strands zip together, a link to Course 6.

Key ideas to carry forward

- Ionic: transfer, lattices. Covalent: sharing. Metallic: electron sea
- VSEPR: electron pairs spread out; lone pairs squeeze angles
- Shape decides whether polar bonds cancel
- Hydrogen bonds explain water's unusual properties

Module 4.4

The mole and stoichiometry

Chemists can't count atoms one by one, so they count them by weighing. The mole is the bridge between the mass you can measure on a balance and the number of particles that actually react.

By the end of this module you can

- Convert between mass, moles and number of particles
- Balance chemical equations and use mole ratios
- Calculate concentrations, yields and limiting reagents

The mole

A mole is a fixed number of particles: six point zero two times ten to the twenty-three, Avogadro's number. It is chosen so that one mole of an element weighs its relative atomic mass in grams. One mole of carbon-twelve weighs twelve grams; one mole of water weighs eighteen grams. To find moles, divide mass by molar mass [Equation 4.4-1]. To find particles, multiply moles by Avogadro's number [Equation 4.4-2].

Balanced equations

Atoms are neither created nor destroyed in a reaction, so equations must balance: the same number of each atom on both sides. Balance by changing the numbers in front of formulas, never the subscripts inside them. For burning methane: one methane plus two oxygen gives one carbon dioxide plus two water [Equation 4.4-3].

The numbers in front are mole ratios. That equation says one mole of methane needs two moles of oxygen. So sixteen grams of methane, one mole, needs sixty-four grams of oxygen, two moles. This is stoichiometry: using an equation as a recipe.

Limiting reagents, yield and concentration

When reactants are not in the exact ratio, one runs out first. It is the limiting reagent, and it decides how much product forms; the other is in excess. Real reactions rarely give the theoretical amount. Percentage yield compares what you got with what the equation predicts [Equation 4.4-4].

Solutions are measured by concentration: moles of solute per litre of solution [Equation 4.4-5], written mol per litre or M. Half a mole of salt in two litres gives a quarter-molar solution. Diluting a solution adds solvent but no solute, so the moles stay the same: concentration times volume before equals concentration times volume after.

Key ideas to carry forward

- $n = m/M$; particles $= n \times N_a$
- Balance with coefficients, never subscripts
- Coefficients are mole ratios
- $c = n/V$; dilution keeps $c_1V_1 = c_2V_2$

Module 4.5

States of matter and gas laws

Solids, liquids and gases differ in how freely their particles move. Gases are the simplest: at ordinary conditions, all gases follow the same law, whatever they are made of.

By the end of this module you can

- Explain the states of matter with the particle model
- Use Boyle's, Charles' and the ideal gas law
- Explain pressure and temperature in kinetic theory

The particle model

In a solid, particles vibrate about fixed positions. In a liquid, they are still close together but slide past one another. In a gas, they are far apart and move fast in random directions, colliding with each other and the container walls. Changing state means overcoming the forces between particles, which takes energy without raising temperature, the latent heat of Module 3.8.

The gas laws

Robert Boyle found in 1662 that, at fixed temperature, a gas's pressure is inversely proportional to its volume [Equation 4.5-1]: squeeze it to half the volume and the pressure doubles. Jacques Charles found that, at fixed pressure, volume is proportional to temperature, as long as temperature is in kelvin [Equation 4.5-2]. Extrapolating Charles' data back to zero volume points to absolute zero, minus two hundred and seventy-three degrees Celsius.

These combine into the ideal gas law: pressure times volume equals moles times the gas constant times temperature [Equation 4.5-3]. R is eight point three one four joules per mole per kelvin. Use pascals, cubic metres and kelvin, and it works for any gas that is not too cold or compressed. At room conditions one mole of any gas fills about twenty-four litres.

Why gases behave this way

Kinetic theory explains the laws. Pressure is the force of countless particles hitting the walls. Squeeze the gas and particles hit the walls more often, so pressure rises. Heat it and particles move faster, hitting harder and more often. Temperature is a measure of the average kinetic energy of the particles [Equation 4.5-4]. Real gases deviate from the ideal law when particles are close enough for their size and attractions to matter, which is how gases eventually liquefy.

Key ideas to carry forward

- Particle spacing and motion distinguish solid, liquid, gas
- Boyle: PV constant. Charles: V/T constant (kelvin!)
- Ideal gas law: PV = nRT
- Temperature measures average particle kinetic energy

Module 4.6

Energy, rates and equilibrium

Some reactions release energy and some absorb it; some are instant and some take years; many never finish but settle into a balance. This module explains all three.

By the end of this module you can

- Distinguish exothermic and endothermic reactions and read energy profiles
- Explain what controls reaction rate, including catalysts
- Describe dynamic equilibrium, Le Chatelier's principle and equilibrium constants

Energy changes

Breaking bonds takes energy; making bonds releases it. If the bonds made are stronger than the bonds broken, the reaction releases energy as heat and is exothermic, with a negative enthalpy change, delta H [Equation 4.6-1]. Burning fuel and respiration are exothermic. If more energy goes in than comes out, the reaction is endothermic, like the cold pack that chills when you squeeze it.

Reaction rates

For particles to react, they must collide with enough energy, called the activation energy, and in the right orientation. Anything that increases successful collisions speeds a reaction up: higher concentration or pressure, higher temperature, larger surface area. Temperature has a big effect, because a small rise greatly increases the fraction of particles with enough energy; a rough rule is that rate doubles for every ten degrees.

Catalysts speed reactions by providing an alternative pathway with lower activation energy, without being used up. Enzymes, the catalysts of life, are covered in Course 6. Industrial catalysts make fertilisers, plastics and fuels.

Equilibrium

Many reactions are reversible. Starting from reactants, the forward reaction slows as they are used up, and the reverse reaction speeds up as products build. Eventually both run at the same rate. Concentrations then stop changing, but both reactions continue: this is dynamic equilibrium.

Le Chatelier's principle says that if you disturb an equilibrium, it shifts to counteract the change. Add more reactant and it shifts towards products. Raise the pressure and it shifts to the side with fewer gas molecules. Raise the temperature and it shifts in the endothermic direction. The Haber process for making ammonia [Equation 4.6-3] is a balancing act of exactly these effects. The equilibrium constant K expresses where the balance lies [Equation 4.6-2]: large K means mostly products.

Key ideas to carry forward

- Exothermic: ΔH negative, heat released. Endothermic: ΔH positive
- Rate rises with concentration, temperature, surface area and catalysts
- Equilibrium is dynamic: equal forward and reverse rates
- Le Chatelier: the system shifts to oppose a change

Module 4.7

Acids, bases and redox

Two kinds of reaction run much of chemistry: acids handing over protons, and redox reactions handing over electrons. Between them they explain vinegar and antacids, rust and batteries, respiration and photosynthesis.

By the end of this module you can

- Define acids and bases as proton donors and acceptors; use pH
- Describe neutralisation and titration
- Identify oxidation and reduction and explain simple cells

Acids and bases

An acid is a proton donor: it gives away a hydrogen ion. A base is a proton acceptor. Hydrochloric acid gives its proton to water to form hydronium ions. Strong acids, like hydrochloric acid, ionise completely; weak acids, like vinegar's ethanoic acid, only partially. Strength is about how completely an acid ionises, not how concentrated it is.

Acidity is measured on the pH scale, minus log base ten of the hydrogen ion concentration [Equation 4.7-1], which you met in Module 2.7. Pure water is pH seven, neutral. Each step down is ten times more acidic. Stomach acid is about pH two; blood is held carefully at seven point four.

Neutralisation and titration

An acid and a base react to form a salt and water: neutralisation [Equation 4.7-2]. A titration measures exactly how much base neutralises a known volume of acid, using an indicator that changes colour at the end point. From the volumes and one known concentration you can find the other, using moles from Module 4.4.

Redox

Oxidation is loss of electrons; reduction is gain of electrons. The mnemonic is OIL RIG: oxidation is loss, reduction is gain. They always happen together: if one substance gives electrons, another takes them [Equation 4.7-3]. Rusting is iron being oxidised by oxygen. Respiration oxidises glucose; photosynthesis reduces carbon dioxide.

A battery separates the two halves of a redox reaction so that the electrons must travel through a wire to get from one to the other, doing useful work on the way. The voltage depends on how strongly each half pulls electrons. Electrolysis runs the process backwards, using electricity to drive a non-spontaneous redox reaction, as in extracting aluminium or charging a phone.

Key ideas to carry forward

- Acids donate protons; bases accept them
- pH = −log[H^+]; each unit is ×10
- Acid + base → salt + water
- OIL RIG: oxidation is loss, reduction is gain of electrons

Module 4.8

Organic chemistry essentials

Organic chemistry is the chemistry of carbon. Carbon's ability to form four bonds and long chains gives millions of compounds, including fuels, plastics, medicines and every molecule of life.

By the end of this module you can

- Explain why carbon forms so many compounds
- Name simple alkanes and recognise the main functional groups
- Describe isomers and polymers, and connect to biological molecules

Why carbon?

Carbon has four valence electrons, so it forms four covalent bonds. Carbon-carbon bonds are strong, so carbon can build chains, branches and rings of almost any length. Add hydrogen, oxygen, nitrogen and a few others, and the possibilities are effectively limitless. More than ninety per cent of known compounds contain carbon.

Hydrocarbons and naming

The simplest organic compounds are hydrocarbons. Alkanes have only single bonds and follow the formula C n H two n plus two [Equation 4.8-1]. Their names count carbons: meth, eth, prop, but, then pent, hex and so on, followed by ane. Methane is natural gas; propane and butane fill gas bottles; octane is in petrol. Alkenes have a carbon-carbon double bond, which makes them more reactive.

Functional groups

A functional group is a cluster of atoms that gives a family its characteristic reactions. Alcohols carry the O H group: ethanol is the alcohol in drinks. Carboxylic acids carry C O O H: ethanoic acid is vinegar. Amines carry N H two. An alcohol and a carboxylic acid combine to form an ester [Equation 4.8-2], the source of many fruit flavours, and fats are esters too.

Isomers have the same formula but different arrangements. Butane and methylpropane are both C four H ten, but with different shapes and boiling points. In drugs, mirror-image isomers can have completely different effects in the body.

Polymers and the molecules of life

Polymers are long chains of repeating units. Polyethylene is thousands of ethene units joined together [Equation 4.8-3]. Life is built from natural polymers: proteins are chains of amino acids, starch and cellulose are chains of glucose, and DNA is a chain of nucleotides. Course 6 picks up from exactly here.

Key ideas to carry forward

- Carbon forms four bonds and long chains
- Alkanes C_nH_{2n+2}; meth, eth, prop, but count carbons
- Functional groups decide reactions
- Polymers are repeating units; proteins, starch and DNA are polymers